

1. General Introduction and Professional Context

1.1 - Professional Context and the FinOps Team

Technip Energies France is the host organization for this project. It belongs to Technip Energies, an engineering and technology group active in project delivery and technology, products and services for the energy sector. At group level, Technip Energies reported revenue of EUR 7.2 billion in 2025 and more than 18,000 employees across 35 countries; these figures do not describe the French entity alone (Technip Energies, 2026b).

Within Technip Energies, Digiteam is part of Strategy, Sustainability and Digital. Infrastructure Design sits within Digiteam's Infrastructure & Operations and Digital Employee Experience branch. According to the author's account, the FinOps team is attached to Infrastructure Design. The team has a manager and a FinOps expert, and works with a Data Engineering consultant based in Morocco and a Power BI consultant based in India. The author works in the team as a FinOps Engineer and reports to its manager. These roles bring together knowledge of cloud spending, data preparation and reporting, but a change to a measure or a visual can require coordination across them.

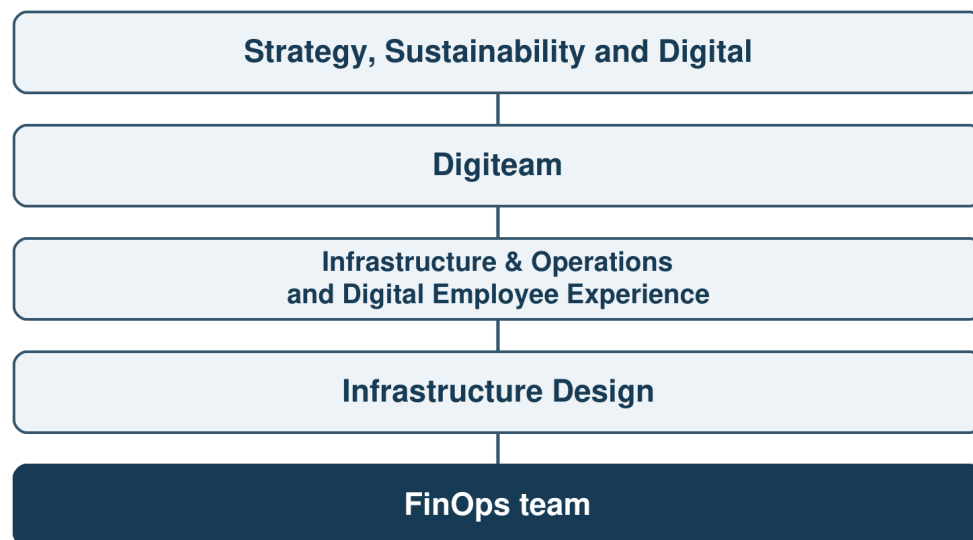


Figure 1.1 - Organizational context of the FinOps team. Source: author's simplified adaptation of internal organizational information dated August 2026; FinOps placement follows the author's professional account. Individual names and unrelated branches are omitted.

FinOps connects technology usage and cost with financial accountability and decisions shared by engineering, finance and business stakeholders (FinOps Foundation, n.d.-c). In this team, cloud consumption data supports analysis of spending, allocation and potential optimization. Cloud Showback is a central reporting product intended to make costs understandable to those responsible for applications and domains, without treating every cost difference as a realized

saving. Its usefulness depends on the data and calculations behind the visual interface as much as on the interface itself. Chapter 2 defines the FinOps concepts and cost measures used in this study.

1.2 - Current Reporting Process and Observed Difficulties

The existing Power BI Cloud Showback report serves FinOps and some management audiences. The author's professional observations identify four difficulties in producing and distributing the information behind it. They are not a measured assessment of the existing system.

Table 1.1 - Difficulties reported in the current Cloud Showback process.

Observed difficulty	Consequence for FinOps
Changes to sources, cost calculations, KPIs such as savings measures, or visuals require coordination with Data Engineering and Power BI contributors. FinOps has limited direct access to the data platform.	The team cannot inspect or implement every routine change itself. Business rules must be agreed across roles, sometimes through several exchanges; the resulting delay has not been measured.
The dashboard is perceived as slow, and some data preparation and calculations take place within Power BI rather than in subject-specific datamarts.	Interactive analysis is less convenient. The respective effects of transformations, the model, queries, visuals, sources and capacity have not been measured.
Application owners need access limited to their applications, while domain managers need a view of the applications in their domain. Interactive access at these scopes is not established for all intended recipients.	For some requests, FinOps performs the analysis and sends a selected Excel extract. A new question may require another manual step.
Blank values, suspected indicator errors and visual defects have been encountered.	Users may question the result, while the cause may lie in source data, a calculation rule or the visual. The frequency and causes of these incidents have not been measured.

These difficulties concern the data path, business definitions, visualization and access decisions together. The consultants' locations describe how work is organized, but distance alone has not been shown to cause the handoffs or delay. Nor does the current distribution process imply that Power BI cannot enforce row-level access. Section 7.2 defines how dashboard performance could be diagnosed; Section 1.3 turns the broader reporting problem into the research question.

1.3 - Problem Statement and Research Question

The difficulties described in Section 1.2 point to one research problem. Cloud-cost reporting depends on a chain of data preparation, quality checks, business definitions and access decisions. When these responsibilities are separated and FinOps has limited ability to inspect or adapt the chain, new reporting needs can require repeated handoffs. A dedicated data platform may make this work more explicit and reusable, but its value must be assessed against what the project actually builds and verifies.

The research question is:

How can a governed FinOps data platform transform FOCUS cost-and-usage data into reliable analytical products for cloud showback, and what operational added value does the resulting system demonstrate?

Here, *added value* has a specific meaning. It concerns the ability to process recurring data automatically, detect and trace data-quality failures, reuse documented measures, prepare datamarts for defined questions and support reporting for authorized audiences. It also concerns the potential to reduce the number of handoffs needed for routine changes. Fewer handoffs do not imply that specialist review, business approval or security controls should disappear. A shorter change time and a lower staffing cost would require a measured comparison with the current process; neither is assumed as a result.

The thesis examines this one question through three connected aspects: the reliability of the data path, the usefulness and adaptability of the analytical products, and the governance of access and operation. These are analytical lenses, not three separate research problems. Section 1.5 distinguishes what the implementation can establish from benefits that would require further measurement.

1.4 - Objectives, Implemented Artifacts and Scope

The project seeks a technical response to the reporting problem, rather than a replacement of the organization's existing Power BI report. It takes FOCUS cost-and-usage files as inputs and develops a reproducible path from those files to documented indicators and a demonstrator dashboard. The intended benefits are clearer responsibility for data and calculations, repeatable processing, easier diagnosis of failures and analytical views prepared for FinOps questions.

The implementation has three linked parts. An independent generator produces fictitious cost-and-usage history from anonymized reference data. A local DuckDB and Streamlit proof of concept tests an end-to-end analytical path. A Databricks implementation on GCP separates DEV and PROD processing, applies a versioned data contract, records operational evidence and prepares Silver, Gold and datamart tables for reporting. Its monthly backfill and incremental Daily workflows address different arrival patterns. The cloud dashboard is a

project demonstrator designed to read these products; it is not evidence that Technip Energies has replaced or adopted the current Power BI Cloud Showback.

The work does not use the organization's detailed billing history as a public research dataset. Synthetic inputs permit reproducible engineering tests, but cannot establish actual savings, allocation accuracy for the company's real applications or the experience of its users. Chapters 3 to 5 explain the method, design and implementation; Chapters 6 to 8 examine the available analytical, technical and cost evidence.

1.5 - Assessment Criteria and Evidence Boundaries

The answer to the research question rests first on whether the data path works as specified. A valid input should pass documented quality rules; an invalid input should stop before publication. Source lineage and run records should identify what was processed. Monthly billing should replace the corresponding provisional Daily period rather than duplicate it, and published totals should reconcile across the relevant layers. Analytical tables should retain clear grains and cost definitions so that the dashboard does not have to reconstruct every business measure from raw records.

Operational added value requires a separate assessment. Automated runs and reusable datamarts can be demonstrated from code and execution records. Whether a proposed view answers a FinOps question can be checked against its definitions and outputs. Restricted interactive access requires tests with representative application-owner and domain-manager identities across the dashboard, exports and direct data access; the presence of access-control features is not such a test. Similarly, a perceived performance improvement requires comparable timings for the existing and proposed reports.

The evidence available for this thesis is uneven. The local proof of concept has recorded reconciliation results for one synthetic monthly dataset. Cloud backfill and validation have been reported, but the corresponding executed versions, run identifiers and control outputs need to accompany any independently verifiable claim. No before-and-after series measures correction time, Power BI latency or the effort of the current cross-team process. Chapter 7 distinguishes implemented controls, observed results and tests still required. Chapter 9 discusses the organizational implications without presenting an unmeasured benefit as a proven outcome.

1.6 - Thesis Structure

Chapter 2 defines the FinOps concepts and cost measures needed to interpret cloud showback. Chapter 3 explains how the intended outcomes led to the dataset, quality controls, analytical preparation and staged implementation. Chapter 4 presents the architecture and data model; Chapter 5 examines the generator, local proof of concept and Databricks implementation.

Chapter 6 analyzes the resulting cost and allocation measures. Chapter 7 evaluates the technical and reporting evidence, including what remains untested in the existing Power BI and cloud environments. Chapter 8 examines the platform's own build and operating costs. Chapter 9 discusses the findings and the possible effect of closer Data Engineering support within FinOps. Chapter 10 answers the research question within the limits of the evidence and identifies further work.

References

FinOps Foundation. (n.d.-c). *FinOps Framework*. <https://www.finops.org/framework/> (accessed 30 September 2026).

Technip Energies. (2026b, 10 March). *Technip Energies publishes its 2025 Annual Report*. <https://investors.technipenergies.com/news-releases/news-release-details/technip-energies-publishes-its-2025-annual-report> (accessed 29 September 2026).